CPE412 Pattern Recognition

Week 6

KNN (K-Nearest Neighbor)

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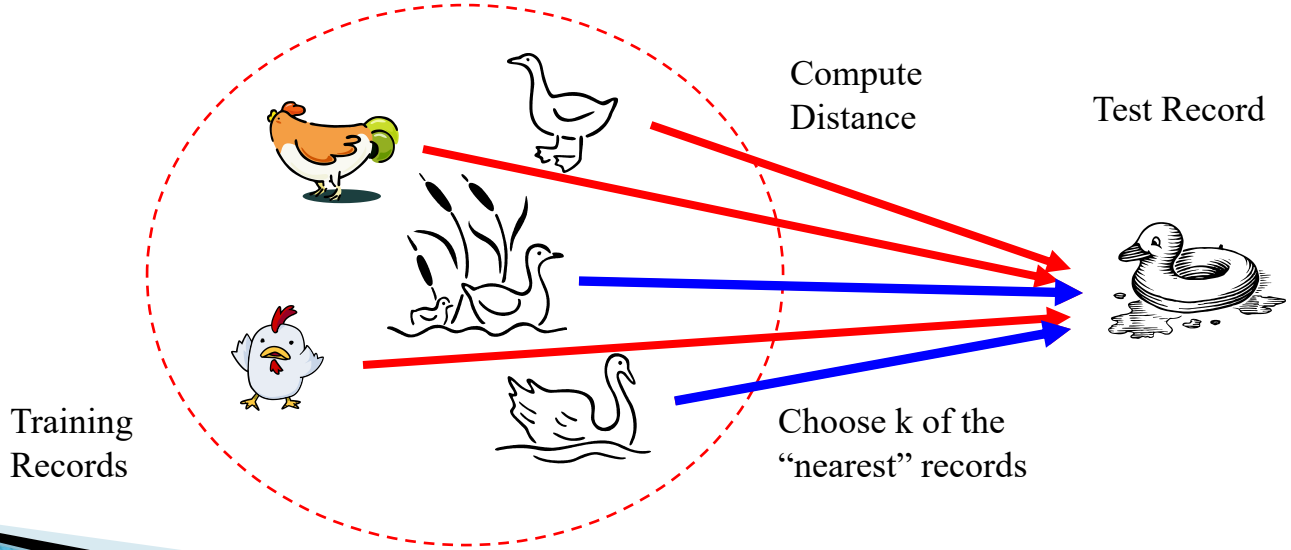
Nearest Neighbor Classifiers

- Basic idea:
  - If it walks like a duck, quacks like a duck, then it's probably a duck



K-Nearest Neighbor

- K-Nearest Neighbor is one of the simplest Machine Learning algorithms based on Supervised Learning technique.
- K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories.
- K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K-NN algorithm.

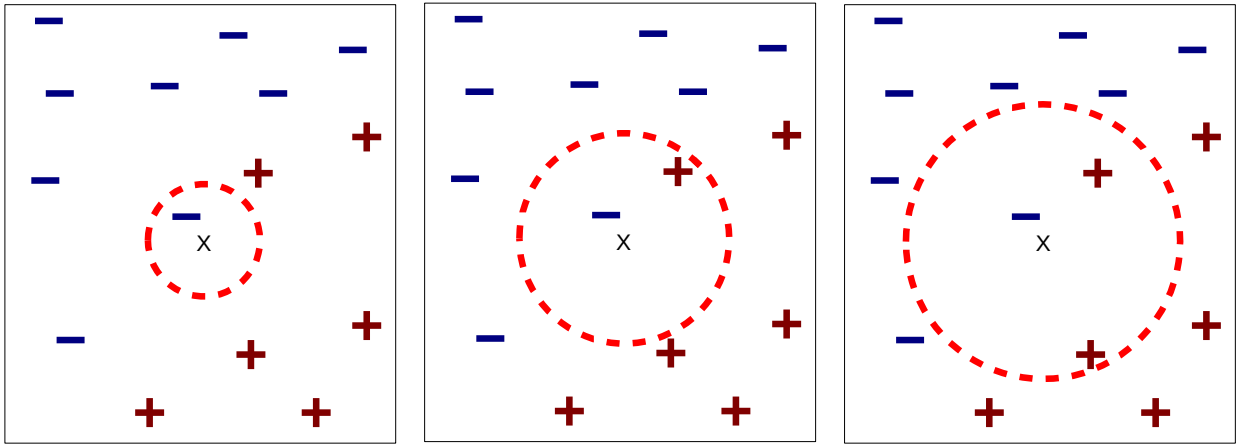
K-Nearest Neighbor

- K-NN algorithm mostly it used for the Classification problems.
- It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.
- KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

K – Nearest Neighbors

- For a given instance T, get the top k dataset instances that are "nearest" to T
  - Select a reasonable distance measure
- Inspect the category(maybe more than 1) of these k instances, choose the category C that represent the most instances.
- Conclude that T belongs to category C
- In practice, k is usually chosen to be odd, to avoid ties.

Definition of Nearest Neighbor



(a) 1-nearest neighbor (b) 2-nearest neighbor (c) 3-nearest neighbor

K-nearest neighbors of a record x are data points that have the k smallest distance to x

How does K-NN work?

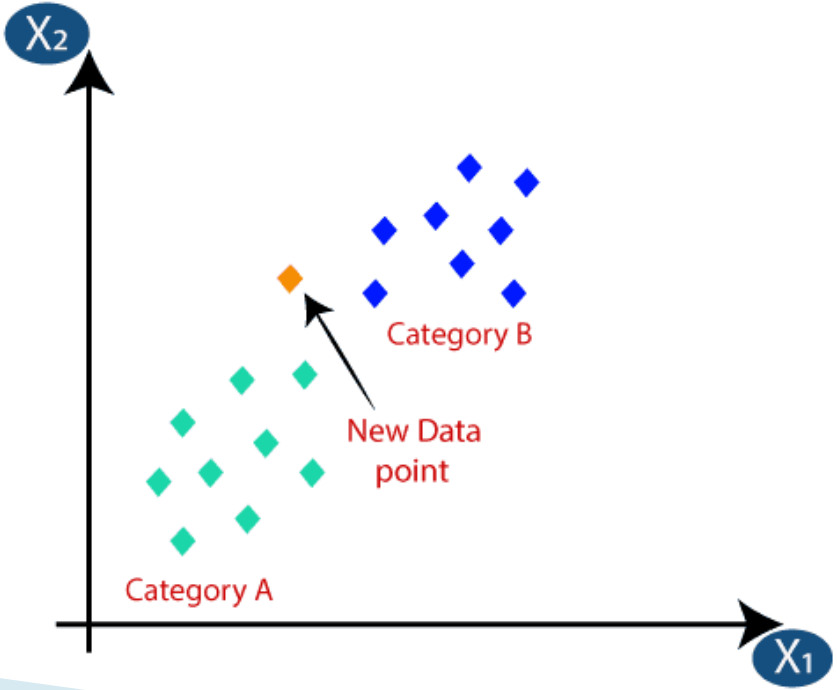
- Step-1: Select the number K of the neighbors
- Step-2: Calculate the Euclidean distance of K number of neighbors
- Step-3: Take the K nearest neighbors as per the calculated Euclidean distance.
- Step-4: Among these k neighbors, count the number of the data points in each category.
- Step-5: Assign the new data points to that category for which the number of the neighbor is maximum.
- Step-6: Our model is ready.

K-Nearest Neighbor

- Advantages of KNN Algorithm:
  - It is simple to implement.
  - It is robust to the noisy training data
  - It can be more effective if the training data is large.
- Disadvantages of KNN Algorithm:
  - Always needs to determine the value of K which may be complex some time.
  - The computation cost is high because of calculating the distance between the data points for all the training samples.

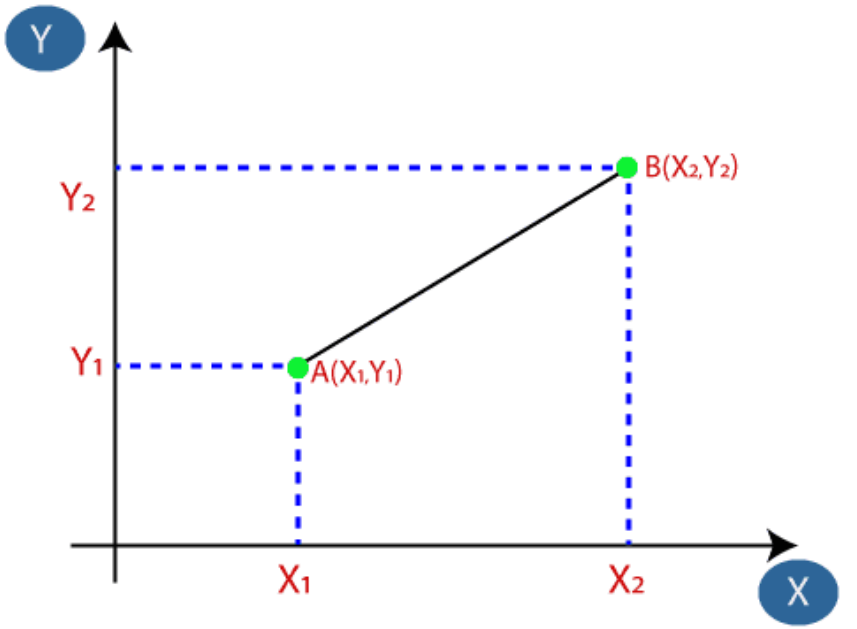
Example1:

- Suppose we have a new data point and we need to put it in the required category. Consider the below image:



Example1:

- Firstly, we will choose the number of neighbors, so we will choose the k=5.
- Next, we will calculate the Euclidean distance between the data points. The Euclidean distance is the distance between two points, which we have already studied in geometry. It can be calculated as:



Euclidean Distance between A_1 and B_2 = $\sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2}$

Example1:

- By calculating the Euclidean distance we got the nearest neighbors, as three nearest neighbors in category A and two nearest neighbors in category B. Consider the below image:

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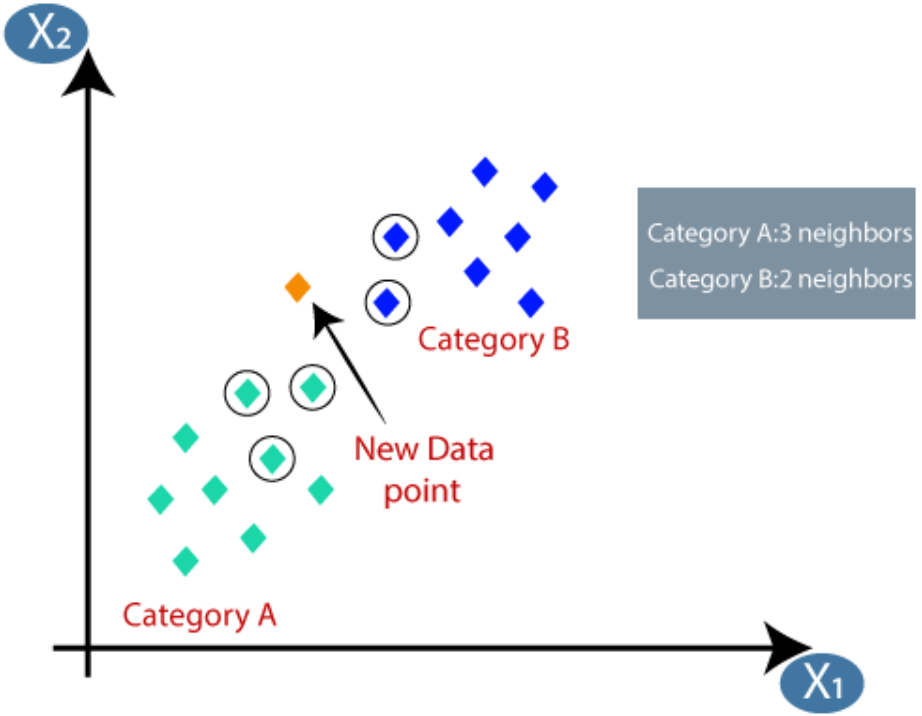
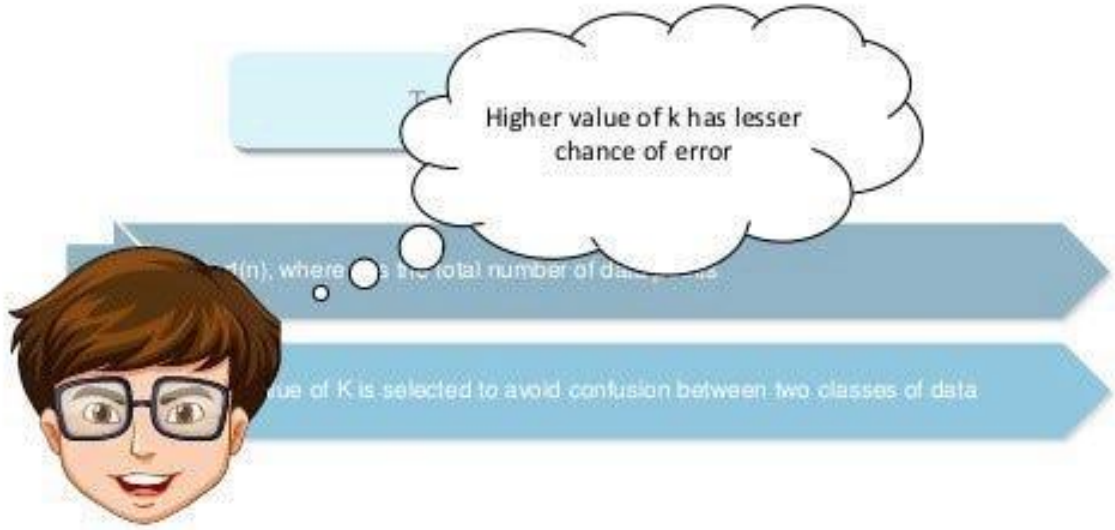


•As we can see the 3 nearest neighbors are from category A, hence this new data point must belong to category A.

How to choose a K value?

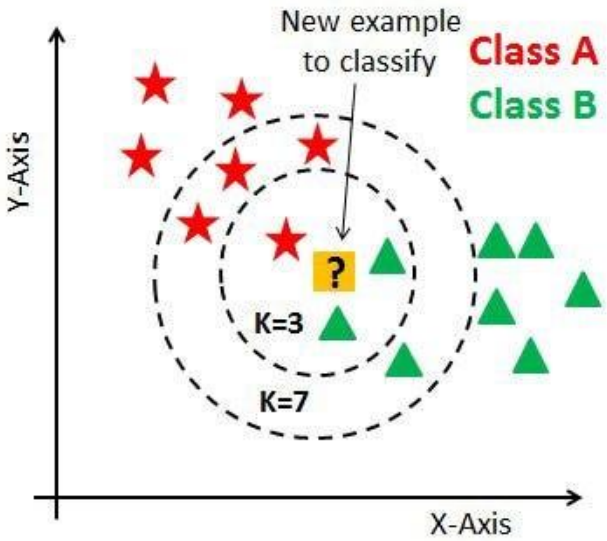


How to choose a K value?

- K value indicates the count of the nearest neighbors. We have to compute distances between test points and trained labels points. Updating distance metrics with every iteration is computationally expensive, and that's why KNN is a lazy learning algorithm.

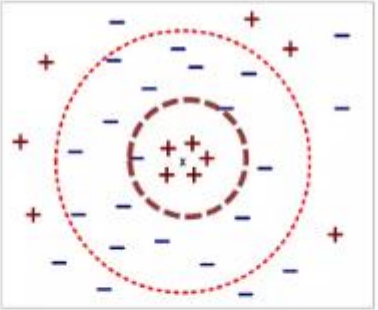
How to choose a K value?

- As you can verify from the image, if we proceed with K=3, then we predict that test input belongs to class B, and if we continue with K=7, then we predict that test input belongs to class A.
- That's how you can imagine that the K value has a powerful effect on KNN performance..



Then how to select the optimal K value?

- If K is too small it is sensitive to noise points.
- Larger K works well. But too large K may include majority points from other classes.



- Rule of thumb is K < sqrt(n), n is number of examples.

Distance Measures

$$Euclidean\ distance:\ d(x,y) = \sqrt{\sum (x_i - yi)^2}$$

$$Squared\ Euclidean\ distance:\ d(x,y) = \sum (x_i - yi)^2$$

$$Manhattan\ distance:\ d(x,y) = \sum |(xi - yi)|$$

Which distance measure to use?

We use Euclidean Distance as it treats each feature as equally important.

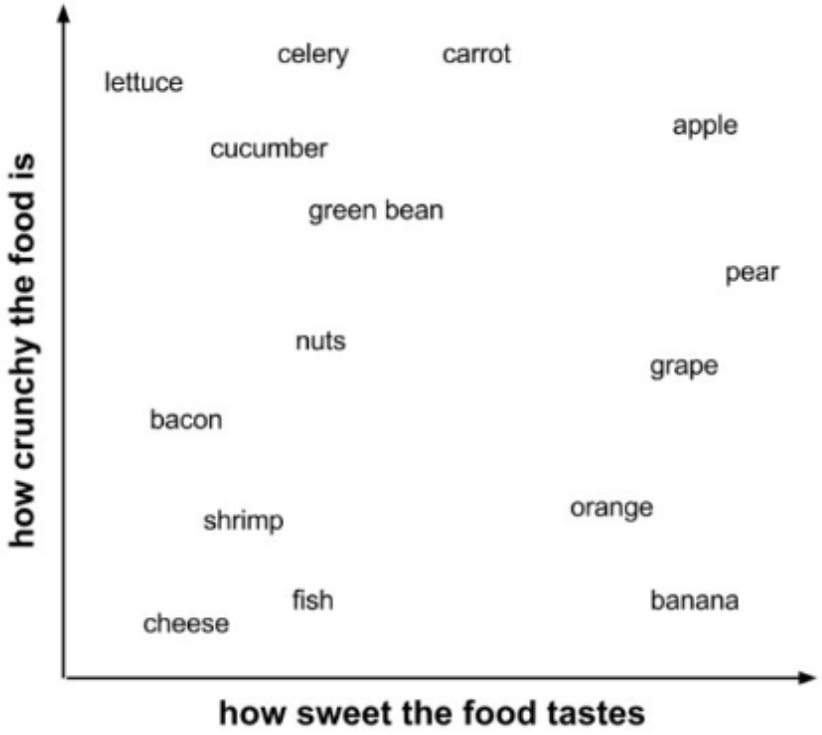
3-KNN: Example(2)

Customer	Age	Income	No. credit cards	Class	Distance from John
George	35	35K	3	No	sqrt $[(35-37)^2+(35-50)^2+(3-2)^2]=15.16$
Rachel	22	50K	2	Yes	sqrt $[(22-37)^2+(50-50)^2+(2-2)^2]=15$
Steve	63	200K	1	No	sqrt $[(63-37)^2+(200-50)^2+(1-2)^2]=152.23$
Tom	59	170K	1	No	sqrt $[(59-37)^2+(170-50)^2+(1-2)^2]=122$
Anne	25	40K	4	Yes	sqrt $[(25-37)^2+(40-50)^2+(4-2)^2]=15.74$
John	37	50K	2	YES	

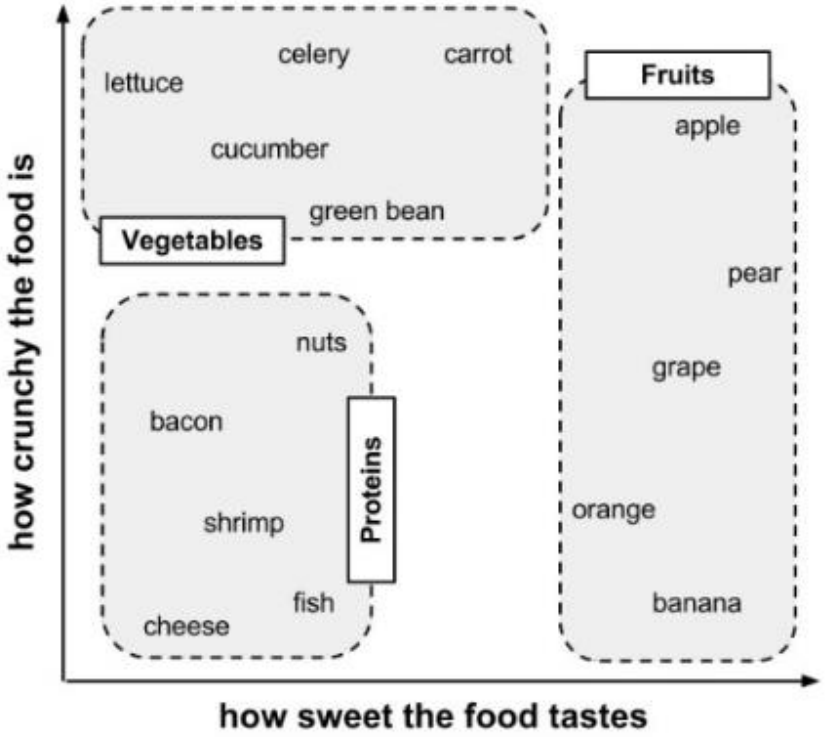
Example 3:

ingredient	sweetness	crunchiness	food type
apple	10	9	fruit
bacon	1	4	protein
banana	10	1	fruit
carrot	7	10	vegetable
celery	3	10	vegetable
cheese	1	1	protein

Example 3:


celery
carrot
lettuce
apple
cucumber
green bean
pear
nuts
grape
bacon
orange
shrimp
fish
banana
cheese
how crunchy the food is
how sweet the food tastes

Example 3:


celery
carrot
lettuce
cucumber
green bean
Vegetables
Fruits
apple
pear
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orange
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nuts
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Proteins
how crunchy the food is
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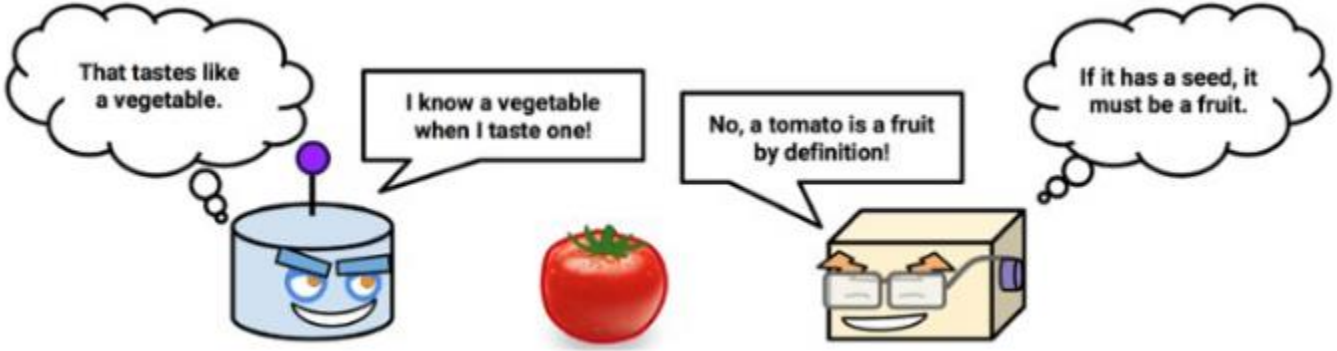
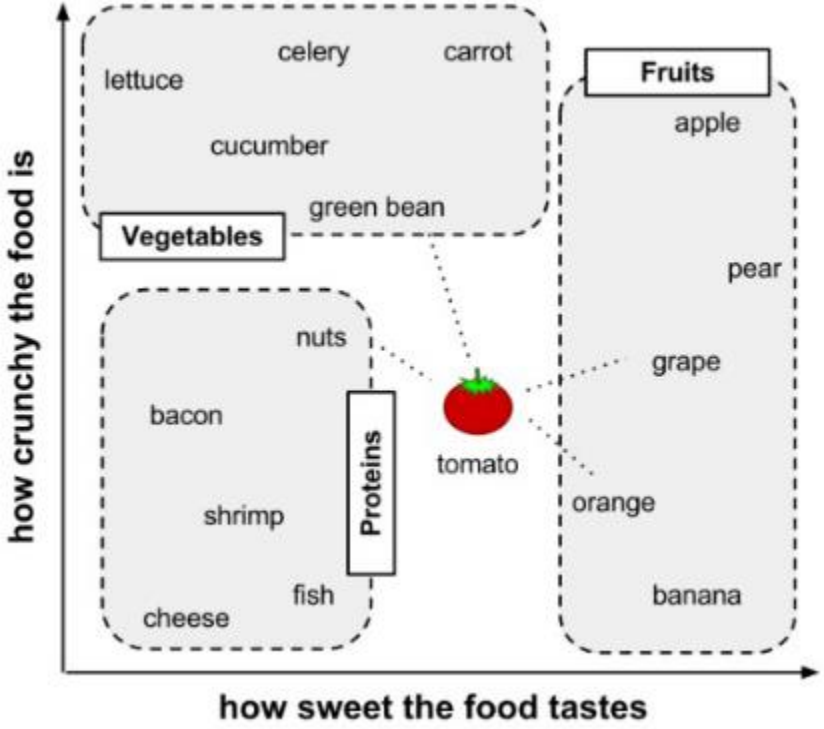
Example 3: Classify me now!

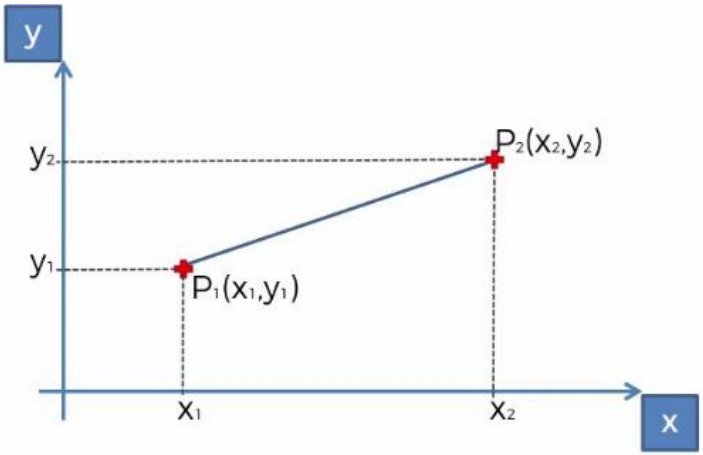


Example 3: Classify me now!



Example 3: Classify me now!(Calculating Distance)

- The distance formula involves comparing the values of each feature. For example, to calculate the distance between the tomato (sweetness = 6, crunchiness = 4), and the green bean(sweetness = 3, crunchiness = 7), we can use the formula as follows:



Euclidean Distance between P_1 and $P_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$\text{dist}(tomato, green\ bean) = \sqrt{(6-3)^2 + (4-7)^2} = 4.2$$

Example 3: Classify me now!(the result)

- In the classifier, we might set k = 4, because there were 15 example ingredients in the training data and the square root of 15 is 3.87.

ingredient	sweetness	crunchiness	food type	distance to the tomato
grape	8	5	fruit	*sqrt((6 - 8)^2 + (4 - 5)^2) = 2.2*
green bean	3	7	vegetable	*sqrt((6 - 3)^2 + (4 - 7)^2) = 4.2*
nuts	3	6	protein	*sqrt((6 - 3)^2 + (4 - 6)^2) = 3.6*
orange	7	3	fruit	*sqrt((6 - 7)^2 + (4 - 3)^2) = 1.4*

Next lecture

- k-means clustering

Thanks ☺